

2. Product Specification and description

2.1. Product Specification

2.1.1. Product Overview

	1) Printing Speed • 24(Color)/ 24(Mono) ppm in A4 (25/25 ppm in Letter) 2) Processor • 533 MHz 3) Printer Language • SPLC, PCL5Ce, PCL6C, PS3, PDF V1.7 4) Memory • 256MB (Standard) / 512MB (Max) 5) Interface • High Speed USB 2.0 • 10/100/1000 BaseTX network connector • 802.11b/g/n wireless LAN (CLP-680DW only) 6) Toner cartridge • Initial : 2K (K)/ 1.5K (CMY) • Sales: • Standard : 2K (K)/ 1.5K (CMY) • High Capacity : 6K (K)/ 3.5K (CMY)
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2.1.2. Specifications



NOTE

Product Specifications are subject to change without notice.

2.1.2.1. General Print Engine

Item		CLP-680ND	CLP-680DW
Engine Speed	Simplex (C/M)	Up to 24/24 ppm in A4 (25/25 ppm in Letter)	Up to 24/24 ppm in A4 (25/25 ppm in Letter)
	Duplex	Up to 12/12 ipm in A4 (12/12 ipm in Letter)	Up to 12/12 ipm in A4 (12/12 ipm in Letter)
FPOT	From Ready (C/M)	As fast as 17/17 sec	As fast as 17/17 sec
	From Sleep (C/M)	As fast as 17/17 sec	As fast as 17/17 sec
Resolution	Optical	600 x 600 dpi	600 x 600 dpi
	Enhanced	Up to 9,600 x 600 effective output	Up to 9,600 x 600 effective output

2.1.2.2. Controller and Software

Item		CLP-680ND	CLP-680DW
Processor		Samsung 533MHz	Samsung 533MHz
Memory	Std.	256 MB	256 MB
	Max.	512 MB	512 MB
Printer Languages		SPLC, PCL5Ce, PCL6C, PS3, PDF V1.7	SPLC, PCL5Ce, PCL6C, PS3, PDF V1.7
Fonts		- PCL : 95 Scalable Fonts (include OCR-A, OCR-B), 1 bitmap - PS : 136 Scalable Fonts	- PCL : 95 Scalable Fonts (include OCR-A, OCR-B), 1 bitmap - PS : 136 Scalable Fonts
Driver	Default Driver	SPL	SPL
	Supporting OS	[WINDOW] XP(32/64bit)/ 2003 Server(32/64bit)/ Vista(32/64bit)/ 2008 Server(32/64bit)/ 7(32/64bit)/ 2008 Server R2	[WINDOW] XP(32/64bit)/ 2003 Server(32/64bit)/ Vista(32/64bit)/ 2008 Server(32/64bit)/ 7(32/64bit)/ 2008 Server R2

Item		CLP-680ND	CLP-680DW
		[Linux] - RedHat Enterprise Linux WS 4, 5 (32/64 bit) - Fedora 5, 6, 7, 8, 9, 10, 11, 12, 13 (32/64 bit) - SuSE Linux 10.1 (32 bit) - OpenSuSE 10.2, 10.3, 11.0, 11.1, 11.2 (32/64 bit) - Mandriva 2007, 2008, 2009, 2009.1, 2010 (32/64 bit) - Ubuntu 6.06, 6.10, 7.04, 7.10, 8.04, 8.10, 9.04, 9.10, 10.04 (32/64 bit) - SuSE Linux Enterprise Desktop 10, 11 (32/64 bit) - Debian 4.0, 5.0 (32/64 bit)	[Linux] - RedHat Enterprise Linux WS 4, 5 (32/64 bit) - Fedora 5, 6, 7, 8, 9, 10, 11, 12, 13 (32/64 bit) - SuSE Linux 10.1 (32 bit) - OpenSuSE 10.2, 10.3, 11.0, 11.1, 11.2 (32/64 bit) - Mandriva 2007, 2008, 2009, 2009.1, 2010 (32/64 bit) - Ubuntu 6.06, 6.10, 7.04, 7.10, 8.04, 8.10, 9.04, 9.10, 10.04 (32/64 bit) - SuSE Linux Enterprise Desktop 10, 11 (32/64 bit) - Debian 4.0, 5.0 (32/64 bit)
		Mac OS X 10.4~10.7	Mac OS X 10.4~10.7
		[UNIX] - Sun Solaris 9,10 (x86, SPARC) - HP-UX 11.0, 11i v1, 11i v2, 11i v3 (PA-RISC, Itanium) - IBM AI 5.1, 5.2, 5.3, 5.4	[UNIX] - Sun Solaris 9,10 (x86, SPARC) - HP-UX 11.0, 11i v1, 11i v2, 11i v3 (PA-RISC, Itanium) - IBM AI 5.1, 5.2, 5.3, 5.4
	WHQL	- SPL/PCL6/PS driver : Windows 2003(32bit), Windows 2003(64bit), Windows Vista(32bit), Windows Vista(64bit), Windows 7(32bit), Windows 7(64bit), Window2000/XP(32/64bit) - XPS driver : Windows Vista(32bit), Windows Vista(64bit), Windwos 7(32bit), Windows 7(64bit)	- SPL/PCL6/PS driver : Windows 2003(32bit), Windows 2003(64bit), Windows Vista(32bit), Windows Vista(64bit), Windows 7(32bit), Windows 7(64bit), Window2000/XP(32/64bit) - XPS driver : Windows Vista(32bit), Windows Vista(64bit), Windwos 7(32bit), Windows 7(64bit)
Wired Network	Protocol	[TCP/IP] - TCP/IPv4/IPv6, HTTP, SNMPv1/v2c/v3, SMTP, IPSec, DNS/WINS, DDNS, DHCP, SSL/TLS, BOOTP, AutoIP, Standard TCP/IP printing, LPR, IPP, UPnP(SSDP), Bonjour, Telnet, WSD, SLP, SetIP, AirPrint, ThinPrint [Others] - HTTPs,IPPs, 802.1x (EAP-MD5, EAP-MSCHAPv2, PEAP, TLS), IPSec	[TCP/IP] - TCP/IPv4/IPv6, HTTP, SNMPv1/v2c/v3, SMTP, IPSec, DNS/WINS, DDNS, DHCP, SSL/TLS, BOOTP, AutoIP, Standard TCP/IP printing, LPR, IPP, UPnP(SSDP), Bonjour, Telnet, WSD, SLP, SetIP, AirPrint, ThinPrint [Others] - HTTPs,IPPs, 802.1x (EAP-MD5, EAP-MSCHAPv2, PEAP, TLS), IPSec

Item		CLP-680ND	CLP-680DW
	Supporting OS	[Windows] - Microsoft Windows XP(32/64bits)/ 2003 Server(32/64bits)/ Vista(32/64bits)/ 2008 Server(32/64bit)/ 7/ 2008 Server R2/ 7/ 2008R2 [Mac] - Mac OS X 10.4~10.7 [Linux] - RedHat 8 ~ 9(32bit)/ Enterprise Linux WS 4, 5(32/64bit) - Fedora Core 1 ~ 7 (32/64bit) - SuSE 8.2 ~ 10.2/ Enterprise Desktop 9~10 (32/64bit) - Madrake 9.2 ~ 10.1 (32/64bit) - Mandriva 2005 ~ 2007 - Ubuntu 6.06 ~ 7.04 - Debian 3.1 ~ 4.0 [Novell] - Netware 5.x, 6.x (TCP/IP Only) [Others] - Unix Sun Solaris 8,9,10(SPARC/86) IBM AIX 5.x (6.1) IBM Linux (System P) HP-U 11i (PA-RISC/IA-64) (11.11/11.20/11.22/11.23/11.31)	[Windows] - Microsoft Windows XP(32/64bits)/ 2003 Server(32/64bits)/ Vista(32/64bits)/ 2008 Server(32/64bit)/ 7/ 2008 Server R2/ 7/ 2008R2 [Mac] - Mac OS X 10.4~10.7 [Linux] - RedHat 8 ~ 9(32bit)/ Enterprise Linux WS 4, 5(32/64bit) - Fedora Core 1 ~ 7 (32/64bit) - SuSE 8.2 ~ 10.2/ Enterprise Desktop 9~10 (32/64bit) - Madrake 9.2 ~ 10.1 (32/64bit) - Mandriva 2005 ~ 2007 - Ubuntu 6.06 ~ 7.04 - Debian 3.1 ~ 4.0 [Novell] - Netware 5.x, 6.x (TCP/IP Only) [Others] - Unix Sun Solaris 8,9,10(SPARC/86) IBM AIX 5.x (6.1) IBM Linux (System P) HP-U 11i (PA-RISC/IA-64) (11.11/11.20/11.22/11.23/11.31)
Application	Anyweb Print	NO	NO
	Installer	YES	YES
	Easy Printer Manager	YES	YES
	Easy Color Manager	YES	YES
	Easy Document Creator	NO	NO
	Net PC Fax	NO	NO
	Direct Printing Utility	YES	YES
	Easy Deployment Manager	YES	YES
	Network Management	Set IP, SWAS 5.0 & SWS 2.0, CounThru 2.5 SWAS Plug-In - Job Accounting, Driver Management Plug-in, Device Map Plug-in	Set IP, SWAS 5.0 & SWS 2.0, CounThru 2.5 SWAS Plug-In - Job Accounting, Driver Management Plug-in, Device Map Plug-in

■ Interface

Item		CLP-680ND	CLP-680DW
Parallel	Option Interface	IEEE 1284b Parallel Option Connector	IEEE 1284b Parallel Option Connector
USB	High Speed USB 2.0	Device 1 Port, Host 1 Port	Device 1 Port, Host 1 Port
Wired Network		10/100/1000 Base TX	10/100/1000 Base TX
Wireless Network		N/A	802.11b/g/n
User Interface	LCD	16 characters x 2 lines	16 characters x 2 lines
	LED	4 (ECO, POWER, NAVI, STATUS)	5 (ECO, POWER, NAVI, STATUS, Wireless)
	Key	7 Keys (3*4 Key, Power, ECO, NAVI, OK, Stop, Demo)	7 Keys (3*4 Key, Power, ECO, NAVI, OK, Stop, WPS)

2.1.2.3. Paper Handling

Item		CLP-680ND	CLP-680DW
Standard Capacity	-	250-sheet Cassette Tray@80g/m² 50-sheet Manual Tray	250-sheet Cassette Tray@80g/m² 50-sheet Manual Tray
Max. Capacity	-	820 sheets @80g/m ²	820 sheets @80g/m ²
Printing	Max. Size	216 x 356 mm (8.5" x 14")	216 x 356 mm (8.5" x 14")
	Min. Size	76 x 127 mm (3.0" x 5.0")	76 x 127 mm (3.0" x 5.0")
Multi-purpose tray	Capacity	- Plain Paper : 50 sheet @80g/m² - Transparency : 10 sheet - Label : 10 sheet - Envelope : 5 sheet	- Plain Paper : 50 sheet @80g/m² - Transparency : 10 sheet - Label : 10 sheet - Envelope : 5 sheet
	Media sizes	A4, Letter, Legal, Oficio, Folio, JIS B5, ISO B5, Executive, A5, Statement, A6, Index Card Stock, Postcard, Envelope Monarch, Envelope No-10, Envelope DL, Envelope C5, Envelope C6, Envelope No 9, Custom [76 x 127mm (3.0" x 5.0") ~ 216 x 356mm (8.5" x 14")]	A4, Letter, Legal, Oficio, Folio, JIS B5, ISO B5, Executive, A5, Statement, A6, Index Card Stock, Postcard, Envelope Monarch, Envelope No-10, Envelope DL, Envelope C5, Envelope C6, Envelope No 9, Custom [76 x 127mm (3.0" x 5.0") ~ 216 x 356mm (8.5" x 14")]
	Media type	Plain Paper, Thin Paper, Thick Paper, Cardstock, Thicker Paper, Hole Punched, Transparency, Pre-Printed, LetterHead, Recycled, Archive, Bond, Label, Envelope, Cotton, Colored, Glossy	Plain Paper, Thin Paper, Thick Paper, Cardstock, Thicker Paper, Hole Punched, Transparency, Pre-Printed, LetterHead, Recycled, Archive, Bond, Label, Envelope, Cotton, Colored, Glossy

Item		CLP-680ND	CLP-680DW
	Media weight	- Supported Weight : 60 ~ 220g/m² - Thin Paper : 60~69g/m² - Plain Paper : 70~89g/m² - Thick Paper : 90~120g/m² - Cardstock : 121~163g/m² - Thicker Paper : 164~220g/m² - Bond Paper: 105~120g/m²	- Supported Weight : 60 ~ 220g/m² - Thin Paper : 60~69g/m² - Plain Paper : 70~89g/m² - Thick Paper : 90~120g/m² - Cardstock : 121~163g/m² - Thicker Paper : 164~220g/m² - Bond Paper: 105~120g/m²
	Sensing	Paper Empty : Yes	Paper Empty : Yes
Standard Cassette Tray	Capacity	250 sheets @80g/m² - Envelope 20 sheets	250 sheets @80g/m² - Envelope 20 sheets
	Media sizes	A4, Letter, Legal, Oficio, Folio, JIS B5, ISO B5, Executive, A5, Statement, A6, Postcard, Envelope Monarch, Envelope No-10, Envelope DL, Envelope C5, Envelope C6, Envelope No 9, Custom[98 x 127mm (3.86" x 5.00") ~ 216 x 356mm (8.5" x 14")]	A4, Letter, Legal, Oficio, Folio, JIS B5, ISO B5, Executive, A5, Statement, A6, Postcard, Envelope Monarch, Envelope No-10, Envelope DL, Envelope C5, Envelope C6, Envelope No 9, Custom[98 x 127mm (3.86" x 5.00") ~ 216 x 356mm (8.5" x 14")]
	Media types	Plain Paper, Thin Paper, Thick Paper, Cardstock, Thicker Paper, Hole Punched, Transparency, Pre-Printed, LetterHead, Recycled, Archive, Bond, Label, Envelope, Cotton, Colored, Glossy	Plain Paper, Thin Paper, Thick Paper, Cardstock, Thicker Paper, Hole Punched, Transparency, Pre-Printed, LetterHead, Recycled, Archive, Bond, Label, Envelope, Cotton, Colored, Glossy
	Media weight	- Supported Weight : 60 ~ 220g/m² - Thin Paper : 60~69g/m² - Plain Paper : 70~89g/m² - Thick Paper : 90~120g/m² - Cardstock : 121~163g/m² - Thicker Paper : 164~220g/m² - Bond Paper: 105~120g/m²	- Supported Weight : 60 ~ 220g/m² - Thin Paper : 60~69g/m² - Plain Paper : 70~89g/m² - Thick Paper : 90~120g/m² - Cardstock : 121~163g/m² - Thicker Paper : 164~220g/m² - Bond Paper: 105~120g/m²
	Sensing	- H/W Install Detect: No - Paper Empty : Yes - Paper Type Detect: No - Paper Size Detect: No	- H/W Install Detect: No - Paper Empty : Yes - Paper Type Detect: No - Paper Size Detect: No
Option Cassette (SCF)	Capacity	520 sheets @80g/m ² (550 sheets @75g/m ²)	520 sheets @80g/m ² (550 sheets @75g/m ²)
	Media sizes	A4, Letter, Legal, Oficio, Folio, JIS B5, ISO B5, Executive, A5, Statement, Custom[98 x 210mm (3.86" x 8.3") ~ 216 x 356mm (8.5" x 14")]	A4, Letter, Legal, Oficio, Folio, JIS B5, ISO B5, Executive, A5, Statement, Custom[98 x 210mm (3.86" x 8.3") ~ 216 x 356mm (8.5" x 14")]
	Media types	Plain Paper, Thin Paper, Thick Paper, Cardstock, Hole Punched, Pre-Printed, LetterHead, Recycled, Archive, Bond, Label, Cotton, Colored, Glossy	Plain Paper, Thin Paper, Thick Paper, Cardstock, Hole Punched, Pre-Printed, LetterHead, Recycled, Archive, Bond, Label, Cotton, Colored, Glossy

Item		CLP-680ND	CLP-680DW
	Media weight	- Supported Weight : 60 ~ 163g/m² - Thin Paper : 60~69g/m² - Plain Paper : 70~89g/m² - Thick Paper : 90~120g/m² - Cardstock : 121~163g/m² - Bond Paper: 105~120g/m²	- Supported Weight : 60 ~ 163g/m² - Thin Paper : 60~69g/m² - Plain Paper : 70~89g/m² - Thick Paper : 90~120g/m² - Cardstock : 121~163g/m² - Bond Paper: 105~120g/m²
	Sensing	- H/W Install Detect : Yes - Paper Empty : Yes - Paper Type Detect : No - Paper Size Detect : Yes	- H/W Install Detect : Yes - Paper Empty : Yes - Paper Type Detect : No - Paper Size Detect : Yes
Output Stacking	Capacity (Face-Down)	100 sheets @80g/m ²	150 sheets @80g/m ²
	Output Full sensing	Paper full Sensor	Paper full Sensor
ADF		N/A	N/A
Duplex	Support	Yes	Yes
	Media sizes	A4, LETTER, FOLIO, OFFICIO, LEGAL	A4, LETTER, FOLIO, OFFICIO, LEGAL
	Media types	Plain, Thick, Hole Punched, Pre-Printed, LetterHead, Recycled, Thin, Cotton, Colored, Glossy	Plain, Thick, Hole Punched, Pre-Printed, LetterHead, Recycled, Thin, Cotton, Colored, Glossy
	Media weight	60~120g/m ²	60~120g/m ²
Non-Printable Area	Other Media	- Top & Bottom: 4.23 mm (0.2") from edge - Left & Right: 3 mm (0.11") from edge	- Top & Bottom: 4.23 mm (0.2") from edge - Left & Right: 3 mm (0.11") from edge
	Envelope	- Top & Bottom: 10 mm (0.4") from edge - Left & Right: 10 mm (0.4") from edge	- Top & Bottom: 10 mm (0.4") from edge - Left & Right: 10 mm (0.4") from edge

2.1.2.4. Reliability and Service

Item	CLP-680ND	CLP-680DW
Printing Volume (SET AMPV)	2,400 sheets / month	2,400 sheets / month
Max. Monthly Duty	40,000 sheets	40,000 sheets

2.1.2.5. Environment

Item		CLP-680ND	CLP-680DW
Acoustic Noise Level(Sound Power/Pressure)	Printing	Less than 52 dBA(Sound Pressure)	Less than 52 dBA(Sound Pressure)
	Standby	Less than 37.0 dBA(Sound Pressure)	Less than 37.0 dBA(Sound Pressure)
	Sleep	Back Ground Level	Back Ground Level
Power Consumption	Ready	Less than 10W	Less than 10W
	AVG.	Less than 450W	Less than 450W
	Max/Peak	1400W	1400W
	Power Save	Less than 1.3W	Less than 2.0W
Dimension (W x D x H)	SET	453 x 420 x 312 mm (17.83 x 16.54 x 12.28 inches)	453 x 420 x 312 mm (17.83 x 16.54 x 12.28 inches)
Weight	SET	19.96 Kg (44 lbs)	19.96 Kg (44 lbs)

2.1.2.6. Consumables

Item		CLP-680ND	CLP-680DW
Toner Cartridge	Model	- Standard yield - CLT-K506S : Black - CLT-Y506S : Yellow - CLT-M506S : Magenta - CLT-C506S : Cyan - High yield - CLT-K506L: Black - CLT-Y506L: Yellow - CLT-M506L: Magenta - CLT-C506L: Cyan	- Standard yield - CLT-K506S : Black - CLT-Y506S : Yellow - CLT-M506S : Magenta - CLT-C506S : Cyan - High yield - CLT-K506L: Black - CLT-Y506L: Yellow - CLT-M506L: Magenta - CLT-C506L: Cyan
	Yield	- Standard yield - Approx. 2,000 pages (K) - Approx. 1,500 pages (CMY) - High yield - Approx. 6,000 pages (K) - Approx. 3,500 pages (CMY)  NOTE Declared cartridge yield in accordance with ISO/IEC 19798.	- Standard yield - Approx. 2,000 pages (K) - Approx. 1,500 pages (CMY) - High yield - Approx. 6,000 pages (K) - Approx. 3,500 pages (CMY)  NOTE Declared cartridge yield in accordance with ISO/IEC 19798.
Waste Toner Container	Model	CLT-W506	CLT-W506
	Yield	Approx. 14,000 pages (Mono) Approx. 3,500 pages (Color)	Approx. 14,000 pages (Mono) Approx. 3,500 pages (Color)

2.1.2.7. Maintenance Parts

Item		CLP-680ND	CLP-680DW
Transfer roller	Part Code	JC95-01638A	JC95-01638A
	Life	100,000 pages	100,000 pages
Fuser unit	Part Code	- JC91-01129A (220V) - JC91-01128A (110V)	- JC91-01129A (220V) - JC91-01128A (110V)
	Life	100,000 pages	100,000 pages
Pick-Up roller	Part Code	JC93-00673A	JC93-00673A
	Life	300,000 pages	300,000 pages
Retard (Reverse) roller	Part Code	JC93-00794A	JC93-00794A
	Life	100,000 pages	100,000 pages
ITB unit	Part Code	JC96-06514A	JC96-06514A
	Life	100,000 pages	100,000 pages

2.1.2.8. Option

Item	Model Name	Support	
		CLP-680ND	CLP-680DW
Memory	ML-MEM370 : 512 MB	Yes	Yes
Option Cassette (SCF)	CLP-S680A	Yes	Yes
Wireless network	-	N/A	Yes (Default)
Hard Disk	-	N/A	N/A
USB to Parallel Cable (IEEE 1284B Parallel connection)	ML-PAR100	Yes	Yes

2.1.3. Model Comparison Table

	Samsung CLP-415NW	Samsung CLP-680ND	HP CP2025dn
Image			
Mono/Color Speed (A4)	18 / 18 ppm	24/ 24 ppm	20/ 20 ppm
Processor	533 MHz	533 MHz	600 MHz
Memory (Std/ Max)	256 MB / 512 MB	256 MB / 512 MB	128 MB / 384 MB
Print Language	SPLC, PCL5Ce, PCL6C, PS3, PDF V1.7	SPLC, PCL5Ce, PCL6C, PS3, PDF V1.7	HP PCL 6, HP PCL 5c, HP postscript3
Paper Input	250 sheets Cassette	250 sheets Cassette 520 sheets Optional Cassette	250 sheets Cassette 250 sheets Optional Cassette
Duplex	N/A	Standard	Standard
Interface	High Speed USB 2.0 10/100/1000 Base TX	High Speed USB 2.0 10/100/1000 Base TX	High Speed USB 2.0 10/100 Base TX
Size (mm)	420 x 426 x 264 mm (16.54 x 16.77 x 10.39 inches)	453 x 420 x 312 mm (17.83 x 16.54 x 12.28 inches)	405 x 484 x 322 mm
Toner Cartridge (K/CMY)	2.5K / 1.8K	• 2K / 1.5K • 6K / 3.5K	3.5K / 2.8K

2.2. System Overview

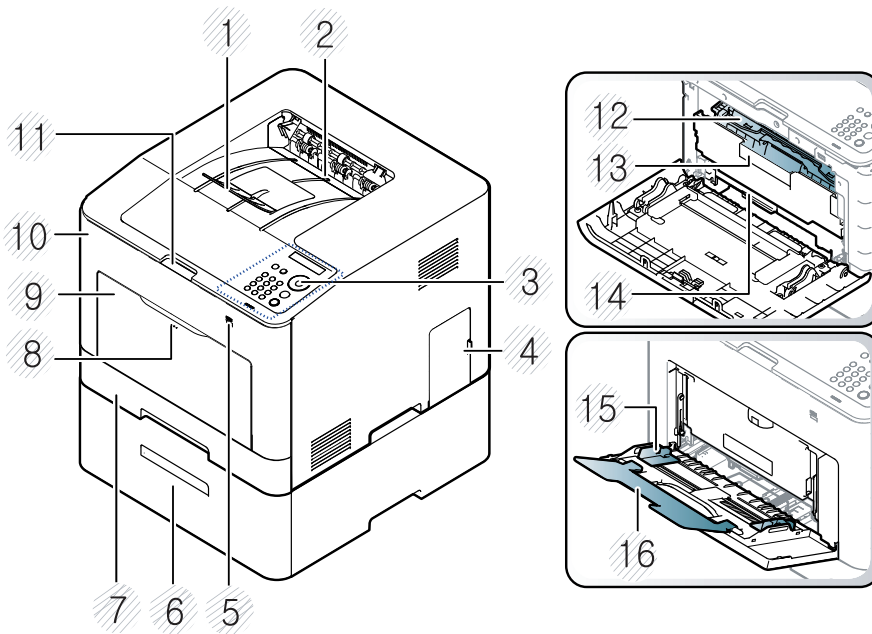
This chapter describes the functions and operating principal of the main component.

2.2.1. Front View



NOTE

- This illustration may differ from your machine depending on your model. There are various types of machine.
- Some features and optional goods may not be available depending on model or country.



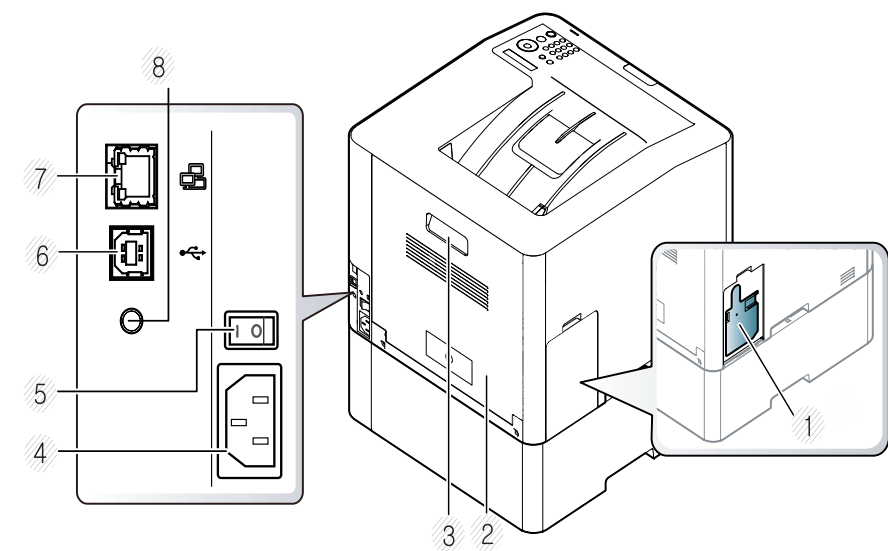
1	Output support
2	Output tray
3	Control panel
4	Control board cover
5	USB memory port
6	Optional tray
7	Tray 1
8	Push-release of Multi-purpose tray
9	Multi-purpose tray

10	Front cover
11	Front cover handle
12	Toner cartridges
13	Toner cartridge handle
14	Intermediate Transfer Belt (ITB)
15	Paper width guides on a multipurpose tray
16	Multi-purpose support tray

2.2.2. Rear View

**NOTE**

- This illustration may differ from your machine depending on your model. There are various types of machine.
- Some features and optional goods may not be available depending on model or country.



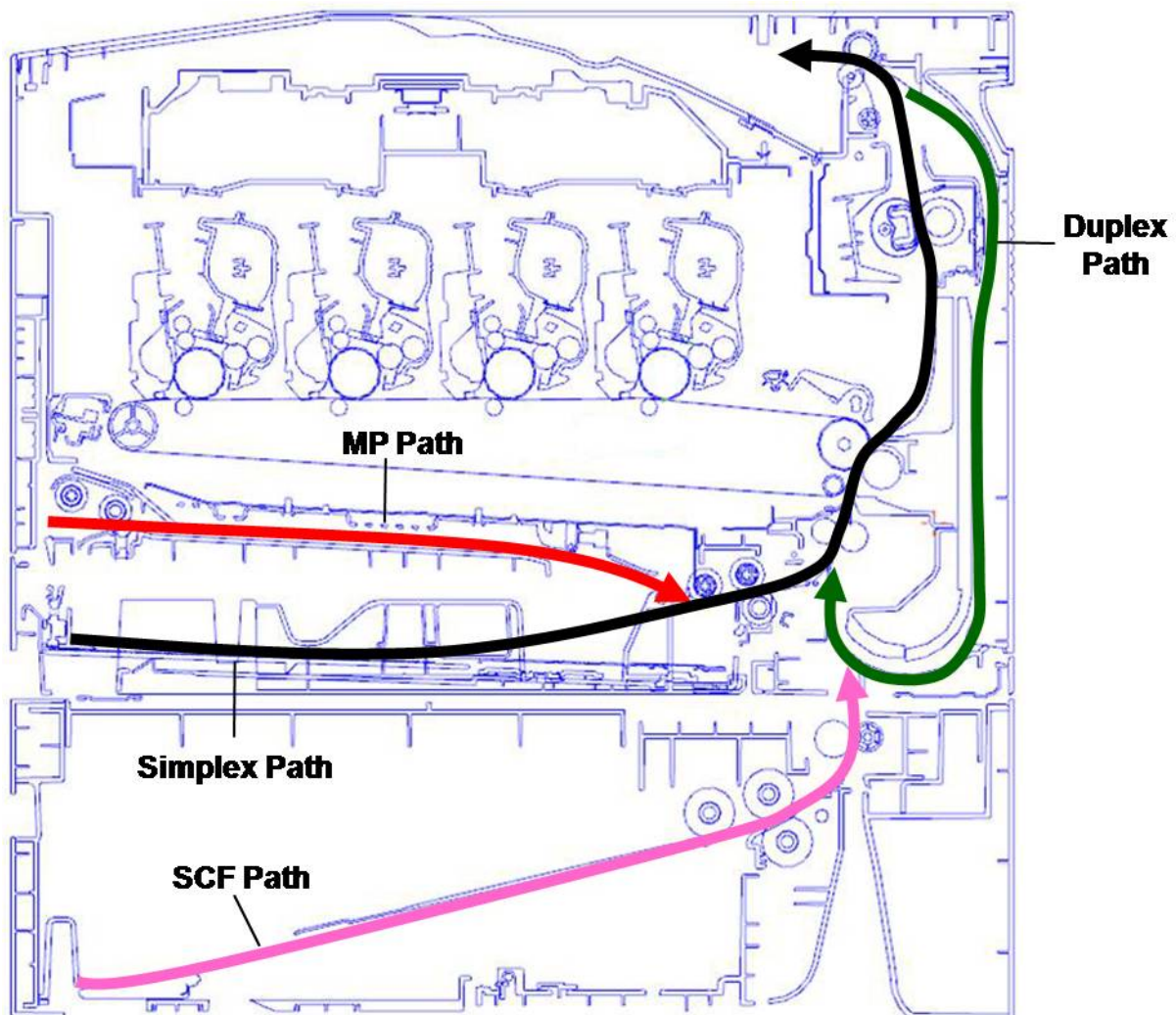
1	Waster toner container
2	Rear cover
3	Rear cover handle
4	Power receptacle
5	Power-switch
6	USB port*
7	Network port
8	5V output port for IEEE 1284 parallel connector*

**NOTE**

* The optional IEEE 1284 parallel connector can be used by plugging in both the USB port and the 5V output port.

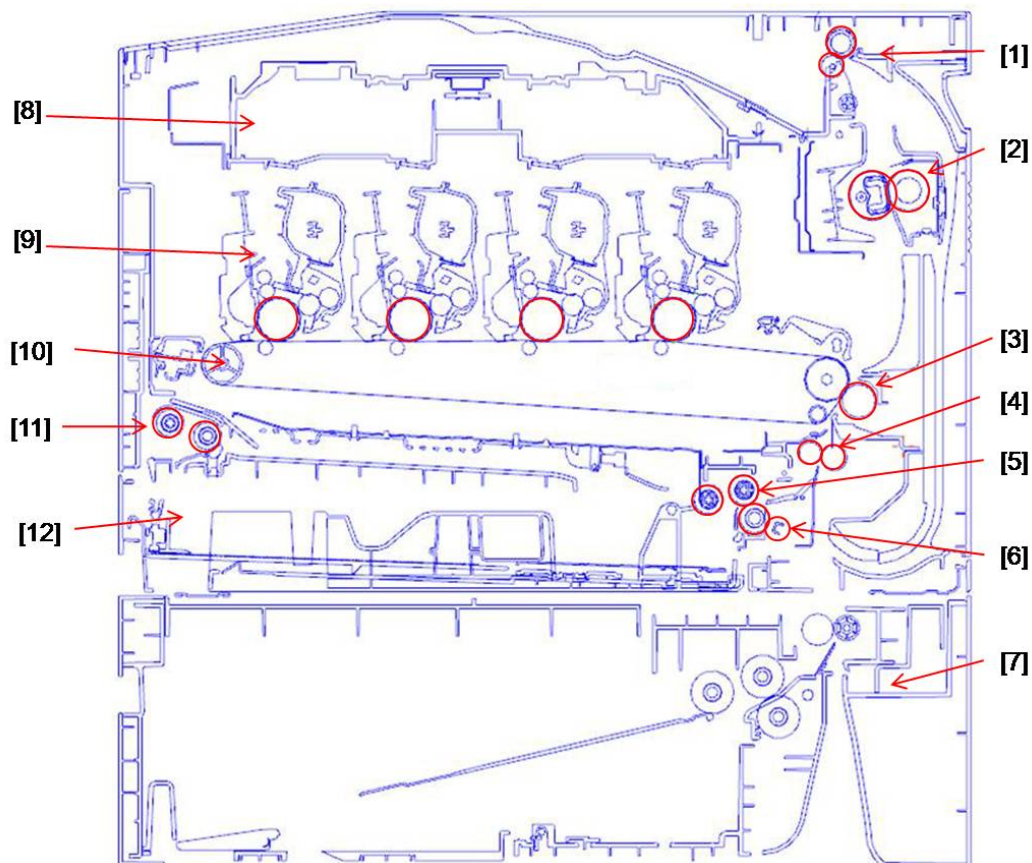
2.2.3. Paper Path

The following diagram displays the path the paper follows during the printing process.



2.2.4. System Layout

This model consists of the engine parts, hardware parts, firmware. The engine parts consists of the mechanical parts comprising Frame, Toner Cartridge, Drive Unit, Transfer roller, Pick up unit, Fuser, Cassette. The hardware parts consists of the main board, SMPS, HVPS board, OPE board, PC interface.



1	Exit Unit
2	Fuser Unit
3	T2 (Second transfer) roller
4	Registration(Regi.) roller
5	Pick up roller
6	Reverse roller
7	SCF (Second Cassette Feeder)

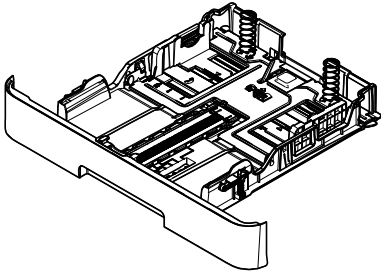
8	LSU
9	Toner Cartridge
10	ITB Unit
11	MP Unit
12	Cassette

2.2.4.1. Feeding Section

It consists of a basic cassette, pick up/forward/retard roller and parts related to paper transferring.

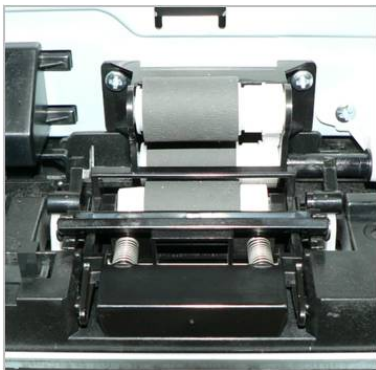
1) Cassette (Tray1)

This model has a cassette type tray. It has a paper existence sensing function, paper storing function, paper arranging function.



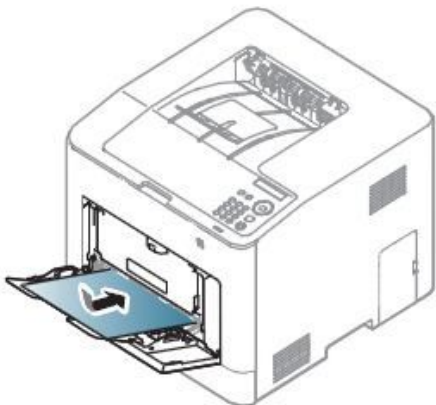
2) Pick up/Forward/Reverse roller

They have functions such as a paper pickup function, driving control function, paper feeding function, and removing electronic static function. Pick up roller is driven by clutch.



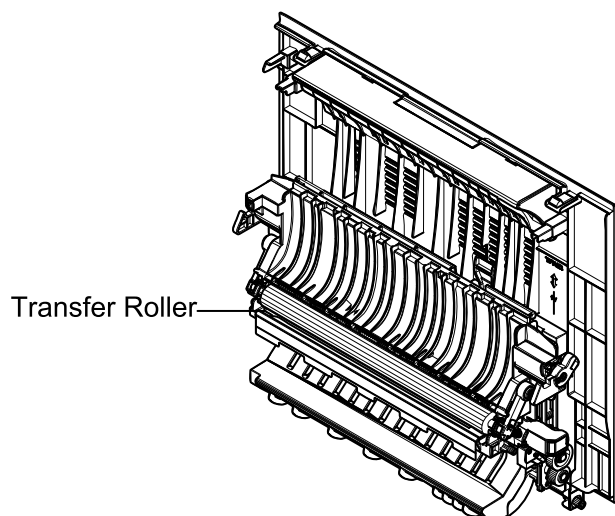
3) MP (Multi-Purpose) Tray

The MP(multi-purpose) tray can hold special sizes and types of print material, such as postcards, note cards, and envelopes. It is useful for single page printing on letterhead or colored paper.



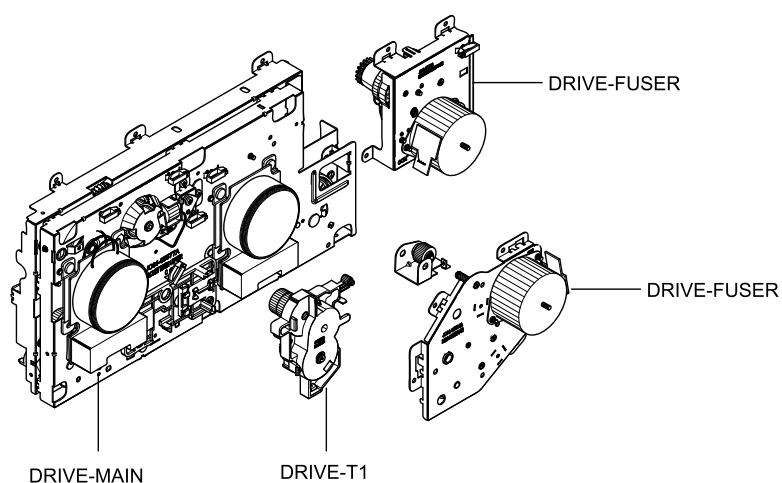
2.2.4.2. Transfer Roller

The transfer roller unit is assembled to the rear cover. The transfer roller delivers the toner of the OPC drum to the paper.



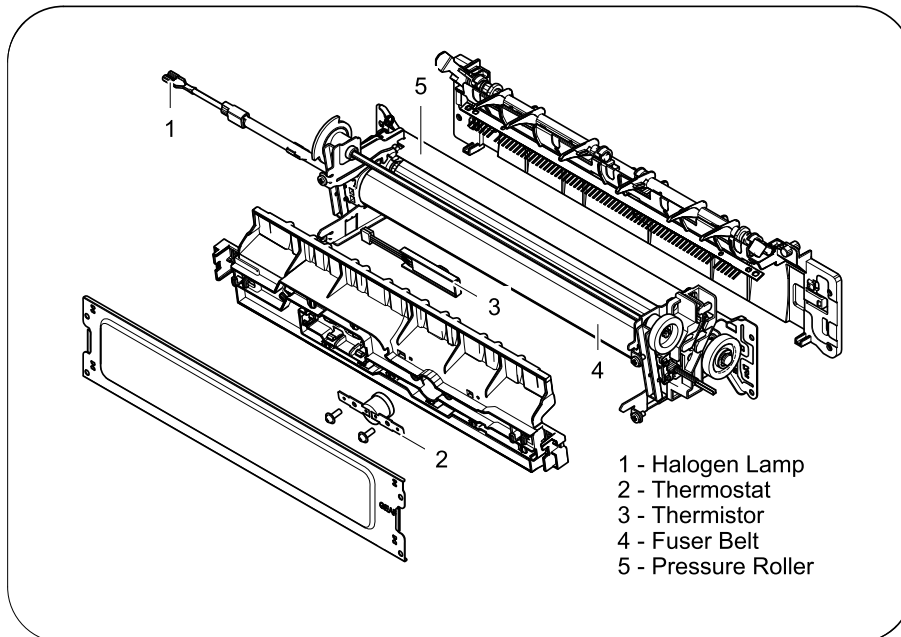
2.2.4.3. Drive Unit

This machine has various drive units. Each drive unit controls the operation for fuser, toner cartridge, duplex, pick up, feeding by using motor and gear train.



2.2.4.4. Fuser Unit

This unit consists of a halogen lamp, a fuser belt, a Thermostat, and Thermistor, etc. It fuses the toner that was transferred by the transfer roller onto the paper, by applying heat and pressure to complete fusing process.



1) Thermostat

When a heat lamp is overheated, a Thermostat cuts off the main power to prevent over- heating.

- Thermostat Type : Non- Contact type Thermostat
- Control Temperature : $195^{\circ}\text{C} \pm 5^{\circ}\text{C}$

2) Thermistor

It is a temperature detecting sensor.

- Temperature Resistance : $7\text{ k}\Omega(180^{\circ}\text{C})$

3) Fusing Belt

The fusing belt gets heat from the halogen lamp and transfer it to toner and paper. The thin fusing belt reduces warming up time and mode changing time.

4) Pressure roller

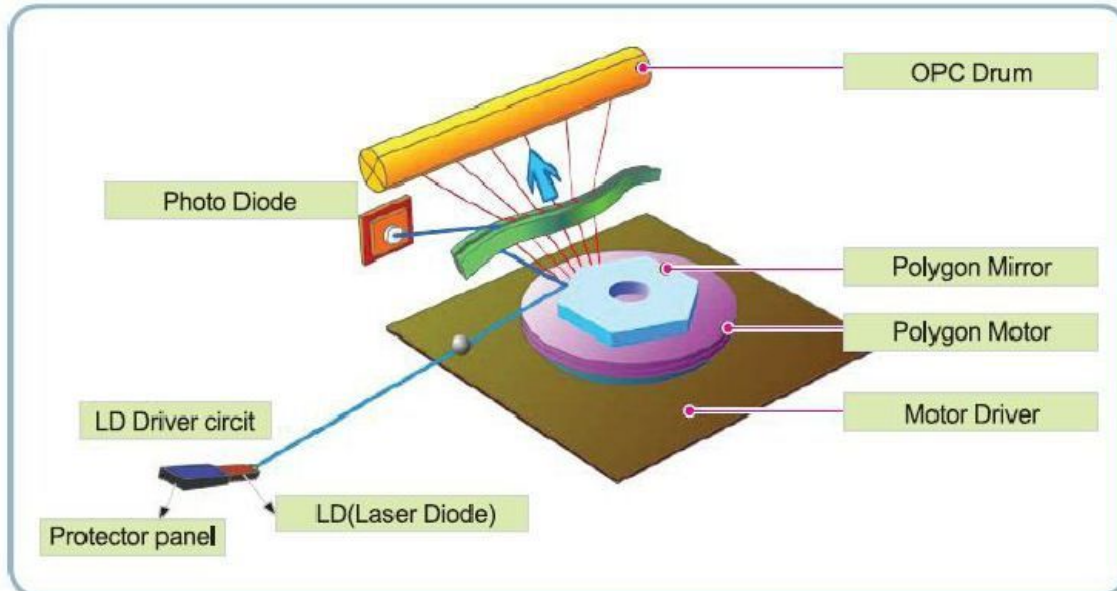
The pressure roller is a rubber roller which ensures proper nip width between the pressure roller and fusing belt. It is driven by the driving system and drives the fusing belt.

5) Halogen Lamp

- Voltage : $120\text{ V} (115 \pm 5 \%) / 220\text{ V} : 230 \pm 5 \%$
- Capacity : $850\text{ Watt} \pm 25\text{ W}$

2.2.4.5. LSU (Laser Scanner Unit)

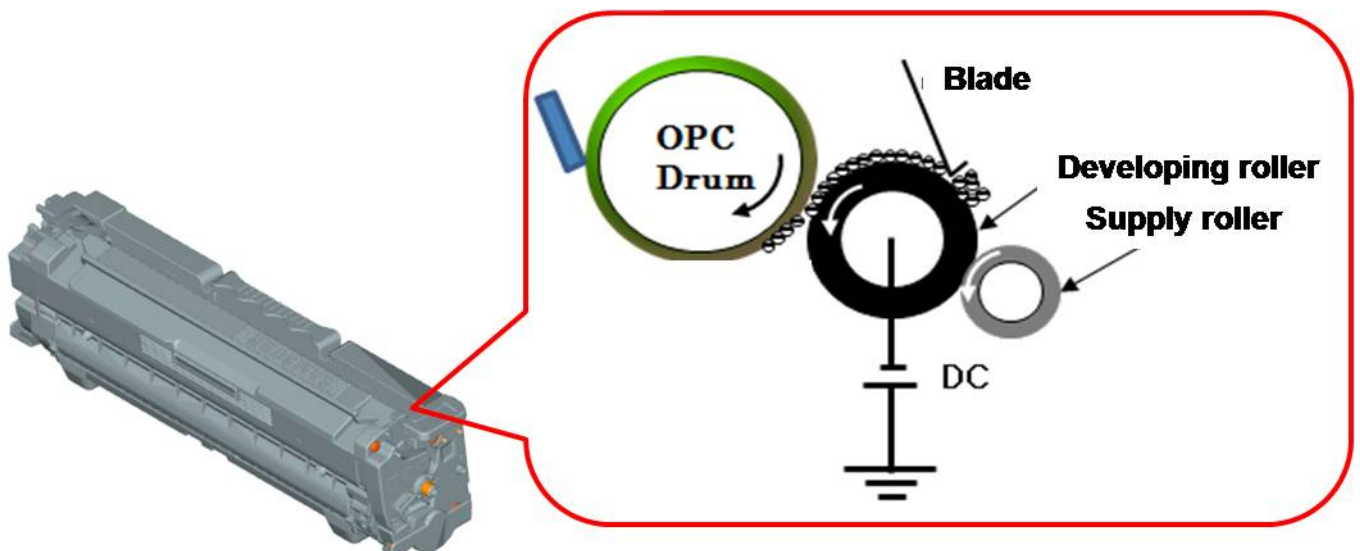
It is the core part of the LBP (Laser Beam Printer) which switches from the video data received to the controller to the electrostatic latent image on the OPC drum by controlling laser beam, exposing OPC drum, and turning principle of polygon mirror. The OPC drum is turned with the paper feeding speed. The HSYNC signal is created when the laser beam from LSU reaches the end of the polygon mirror, and the signal is sent to the controller. The controller detects the HSYNC signal to adjust the vertical line of the image on paper. In other words, after the HSYNC signal is detected, the image data is sent to the LSU to adjust the left margin on paper. The one side of the polygon mirror is one line for scanning.



2.2.4.6. Toner Cartridge

By using the electronic photo process, it creates a visual image. In the toner cartridge, the OPC unit and the developing unit are in a body. The OPC unit has OPC drum and charging roller, and the developing unit has toner, supply roller, developing roller, and blade.

- Developing Method : Non magnetic single component non-contacting method
- Toner : Non magnetic single component pulverized type toner
- The life span of toner (ISO 19798 pattern / A4 standard)
 - Initial toner : 2K (K) / 1.5K (CMY)
 - Sales toner :
 - Standard yield : 2K (K) / 1.5K (CMY)
 - High yield : 6K (K) / 3.5K (CMY)
- OPC Cleaning : Collect the toner by using cleaning blade
- Handling of wasted toner : Collect the wasted toner in the cleaning frame by using cleaning blade
- Classifying device for toner cartridge: ID is classified by CRUM

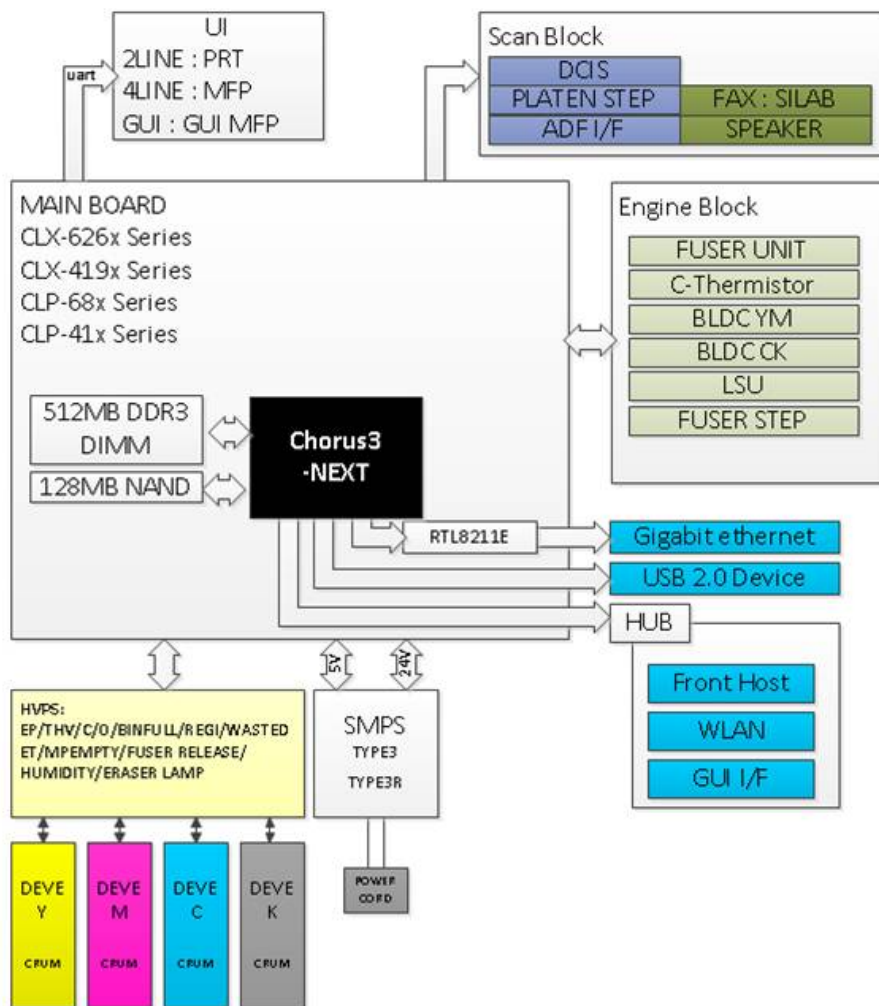


2.2.5. Hardware configuration

The CLP-680 series Electrical Circuit System consists of the following:

- Main board (System board)
- OPE board
- SMPS board
- HVPS board
- HUB board

Diagram of the CLP-680 series Electrical Circuit



CLP-68x series has a system board of integrated engine controller and video controller.

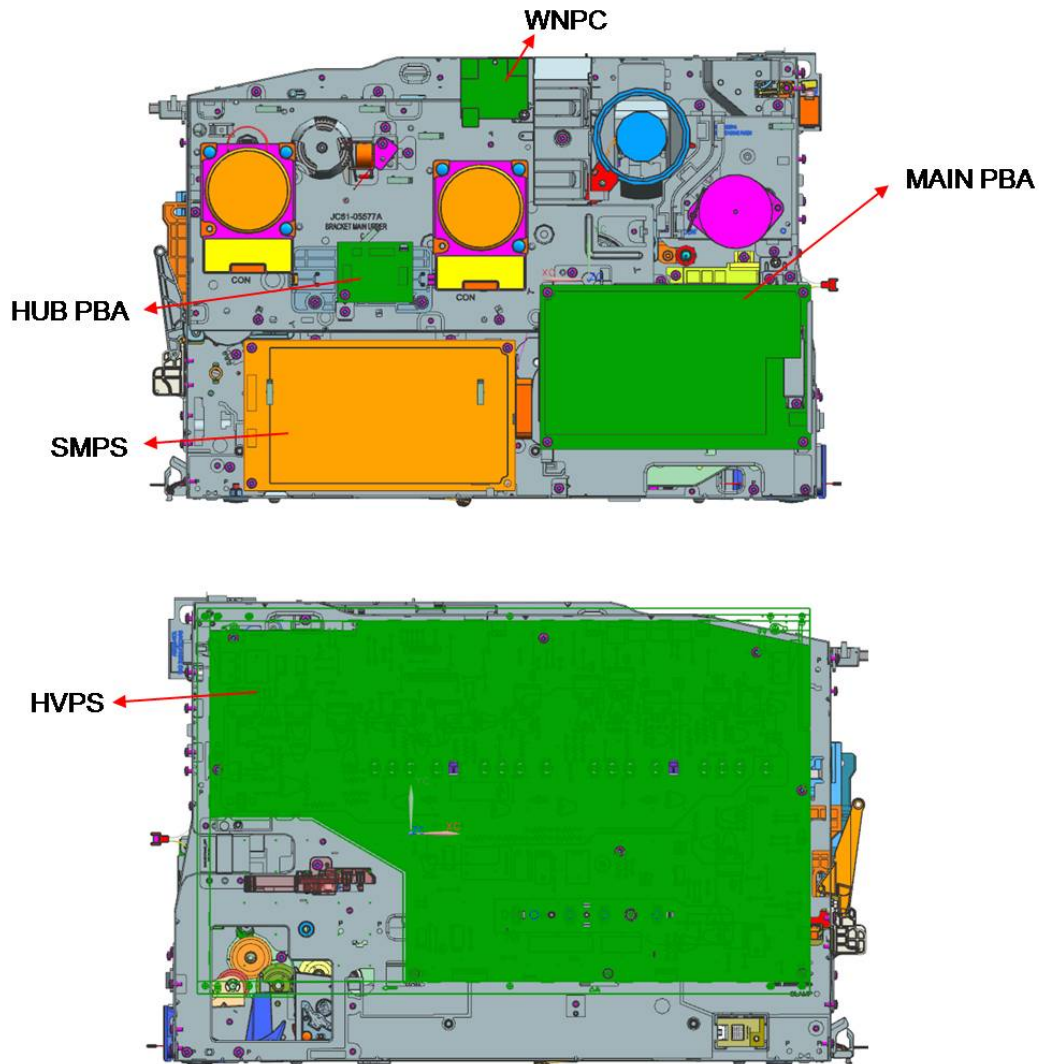
The engine controller controls all modules required to print, that is, LSU, HVPS/SMPS, Fuser, Motor etc. It communicates with the video control block inside CPU for printing. And it has the interface for all video sync signal to print out the video data.

The video controller receives print data from the host through network or USB Port. It takes this information and generates printable video bitmap data.

The main board is adopted 533 MHz C3N CPU that is integrated with engine/ video/ scan controller. It uses DDR3 256MByte.

Circuit board locations

The following diagrams show the locations of the printer circuit boards:



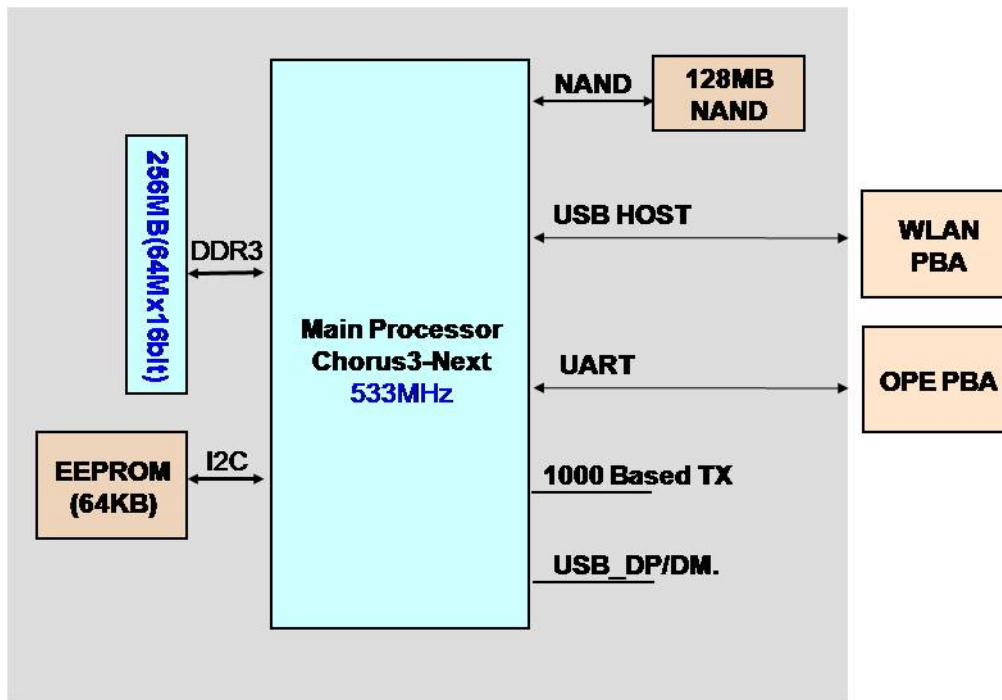
2.2.5.1. Main board

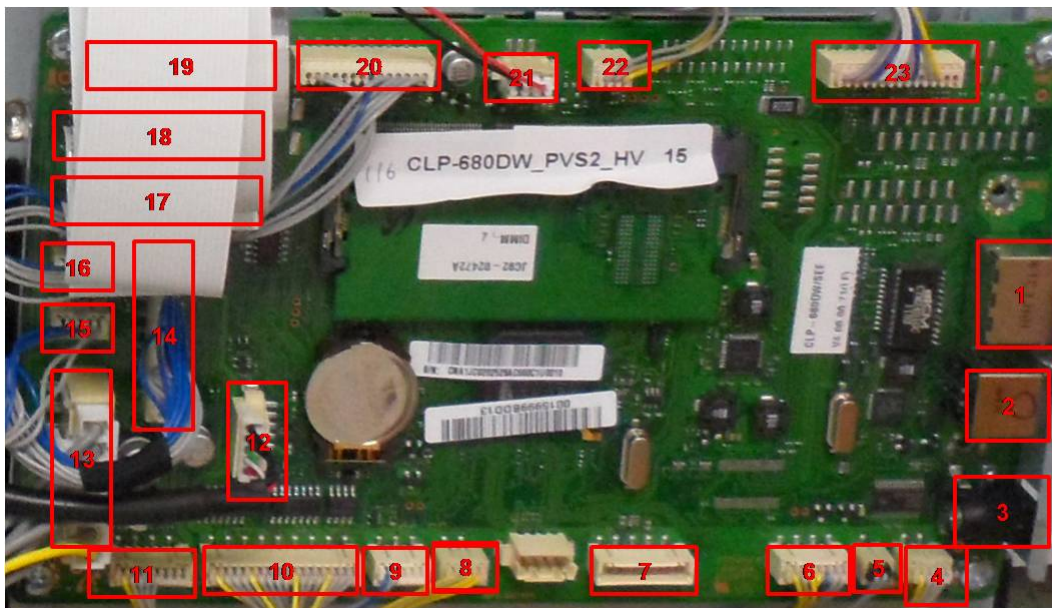
Chorus3–Next(C3N) chip is adopted as the main processor. Its process speed is 533 MHz. It is integrated engine controller, video controller, scan controller.

DDR3 256MB is adopted for high speed data processing. Boot adopted the 128MB NAND Flash.

USB is the embedded type and wired network supports 1000M full duplex.

[Main board diagram]



[Main board image]

- Connection**

1	NETWORK connector
2	USB Device connector
3	1284 I/F connector
4	Feed Step Motor connector
5	Regi. Clutch connector
6	Pick Up Clutch connector
7	SCF connector
8	SMPS Fan T1 DC motor connector
9	MP Empty, Cassette Open connector
10	ITB Joint connector
11	Fuser CONT. connector
12	HUB I/F connector

13	SMPS connector
14	BLDC Motor connector
15	DEVE Home sensor/solenoid connector
16	OPC Home sensor connector
17	HVPS2 connector
18	HVPS1 connector
19	LSU connector
20	OPE connector
21	Fuser Fan connector
22	Thermistor connector
23	Exit Motor connector

- Information**

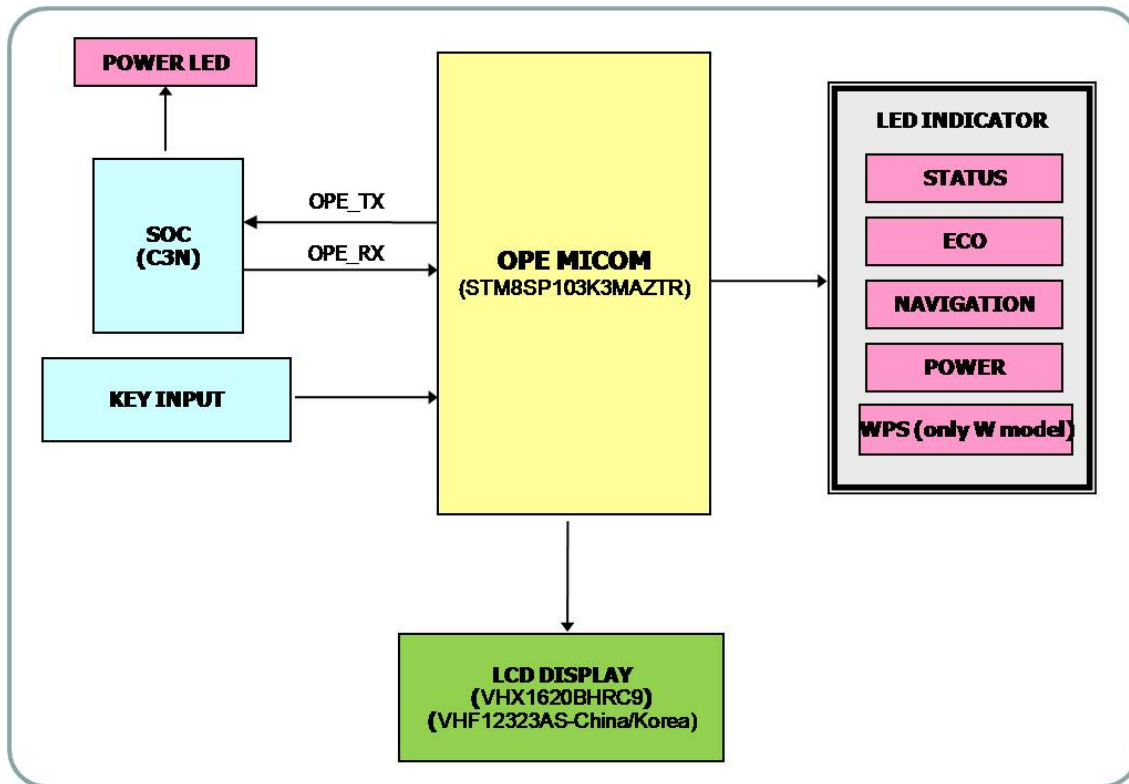
- Part Code
 - JC92-02528A : CLP-680DW
 - JC92-02528B : CLP-680ND

2.2.5.2. OPE Controller

The OPE controller is composed of an MICOM(STM8SP103K3MAZTR),STATUS LED, ECO LED, POWER LED, NAVIGAION LED, WPS LED(CLP-680DW only), 2LINE LCD, KEY

The OPE controller communicates with main controller via UART. The power LED is controlled by the main board.

[OPE controller diagram]



[OPE controller image]



- **Information**
 - Part Code : JC92-02520A~D
 - Part Name : PBA-OPE

- **Connection**

1	Interface Connector to Main board
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2.2.5.3. Wireless LAN board

The Wireless LAN Module supports 802.11b/g/n. It communicates with video controller via USB.



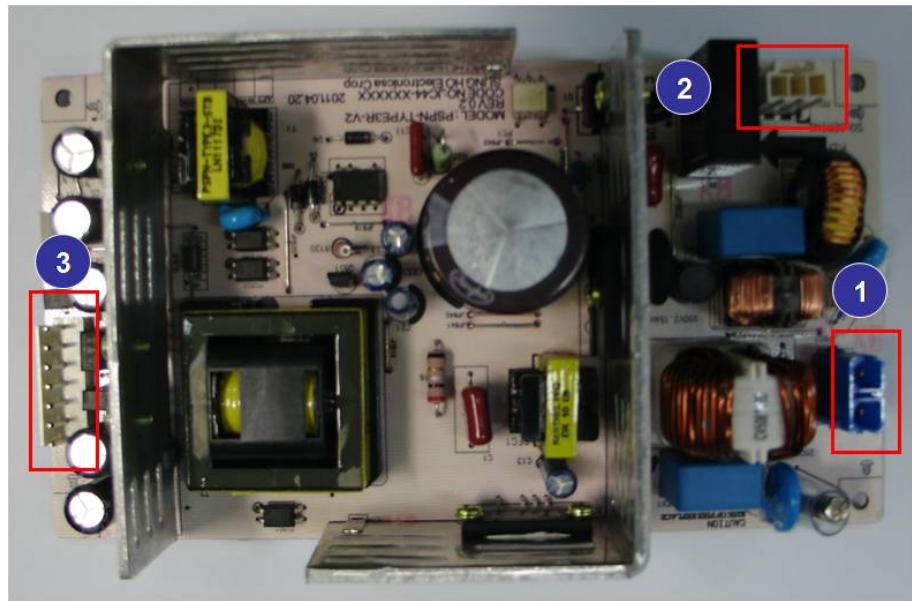
- **Information**
 - Part Code : JC92-02510A
 - PBA Name : PBA-WNPC

- **Connection**

1	Interface Connector to System board (JC39-01858A/ JC39-01803A/ JC39-01853A, HARNESS-WLAN)  NOTE The HARNESS-WLAN differ from your machine depending on your model.
---	--

2.2.5.4. SMPS board

The SMPS (Switching Mode Power Supply) Board supplies electric power to the Main Board and other boards through a Main Controller. The voltage provided includes +24V from a 110V/220V power input.



• Specification

- 1) AC 110V (90V ~ 135V)
- 2) AC 220V (180V ~ 270V)
- 3) Input Current: 10.0A (110V) / 8.0A (220V)
- 4) Output Power: 161.5W
 - DC 5V : 17.5W
 - DC 24V : 144W

• Information

	110V	220V
Part Code	JC44-00222A	JC44-00223A
PBA Name	SMPS V1	SMPS V2

• Connection

1	AC_Inlet
2	FUSER_AC Output
3	OUTPUT_DC

- **Input / Output connector**

- AC Input Connector(CON1)

PIN ASSIGN	PIN NO	Description
1	AC_L	AC Input
2	AC_N	

- AC Input Connector(CON2)

PIN ASSIGN	PIN NO	Description
1	AC_L	AC Output for Heater Controller
2	AC_N	

- DC Output Connector (CON3)

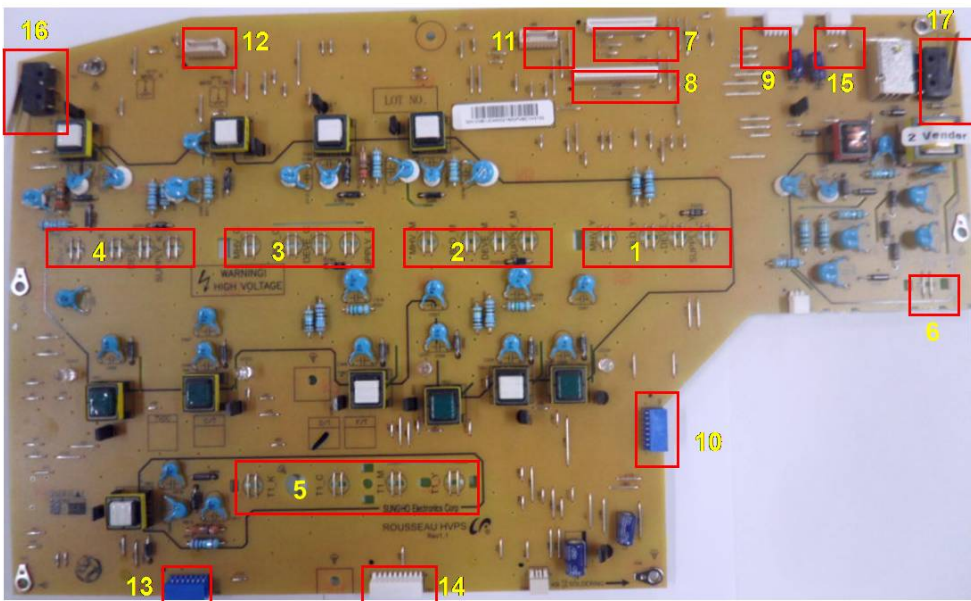
PIN ASSIGN	PIN NO	Description
1	5V	+5V Power
2	GND	Ground
3	24V1	+24V Power
4	GND	Ground
5	24V2	+24V Power
6	GND	Ground
7	24V3	+24V Power

- Signal Connector (CON4)

PIN ASSIGN	PIN NO	Description
1	GND	Ground
2	24V_ON_OFF	24V_ON_OFF
3	Relay on	Fuser Relay
4	24VS	Photo Triac Bias
5	Fuser On	Fuser On

2.2.5.5. HVPS board

The HVPS board generates high-voltage channels which includes MHV, DEV, Blade, SUP, THV1, THV2.



- Connection**

1	MHV/DEV/BLD/SUP Y
2	MHV/DEV/BLD/SUP M
3	MHV/DEV/BLD/SUP C
4	MHV/DEV/BLD/SUP K
5	THV1 YMCK
6	THV2
7	Main-HVPS I/F #1
8	Main-HVPS I/F #2
9	Paper exit, Outbin full
10	WTB interface
11	Crum Y,M
12	Crum C,K
13	Ambient, Humidity
14	Paper Regi
15	Fuser Release
16	Front cover open Switch
17	Rear cover open Switch

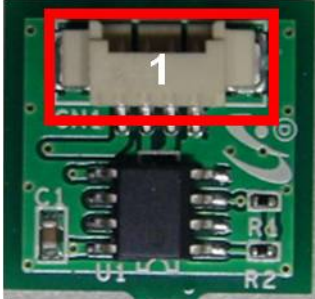
- Information**

- Part Code : JC44-00216A
- PBA Name : HVPS

2.2.5.6. ITB EEPROM PBA

The ITB EEPROM PBA stored the ITB serial number and adjustment information like a S-Gain, ITB life.

It consists of the 4-pin connector, EEPROM, resistor, cap and is connected to the main board. It communicates with main board.



- Information**

- Part Code : JC92-02531A
- PBA Name : PBA-ITB EEPROM

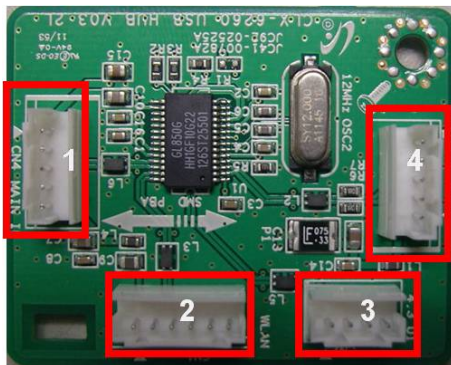
- Connection**

1	Interface Connector to System Board (JC39-01814A, HARNESS-ITB)
---	---

2.2.5.7. HUB PBA

The HUB PBA connects the USB block of the main board and 3 external interface modules.

It consists of 4 connector and Crystal.Hub controller chip.



- Information**

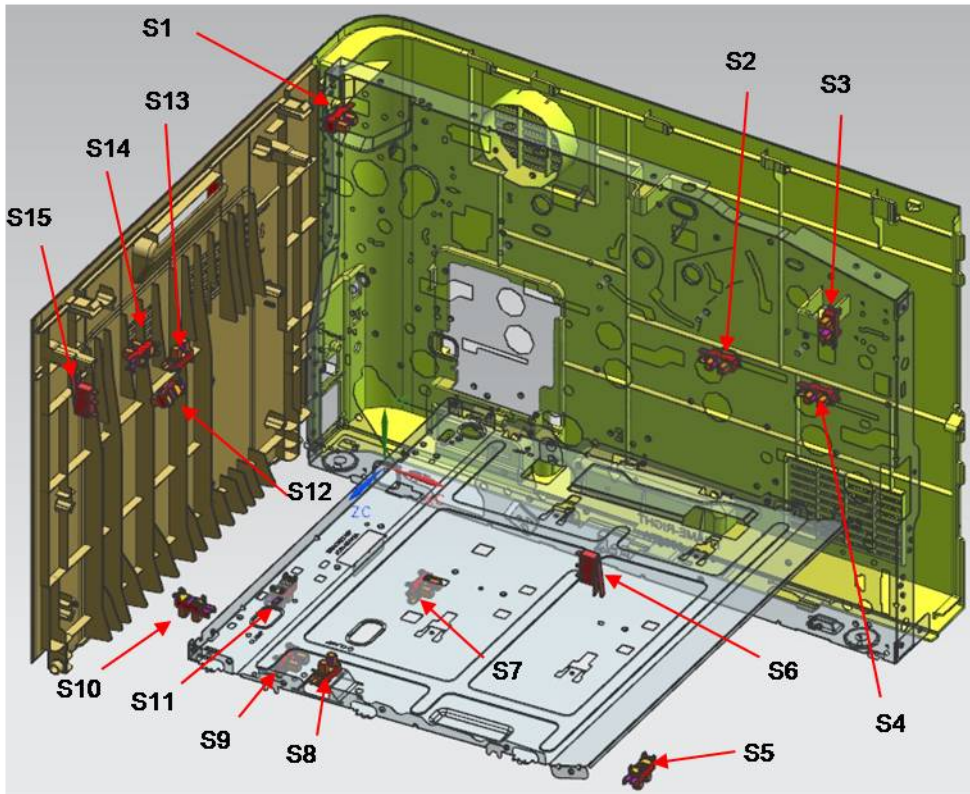
- Part Code : JC92-02525A
- PBA Name : PBA-HUB

- Connection**

1	HUB PBA to Main PBA connector
2	HUB PBA to WLAN connector (CLP-680DW only)
3	HUB PBA to 4.3 inch GUI connector (CLX-6260FW only)
4	HUB PBA to USB Host connector

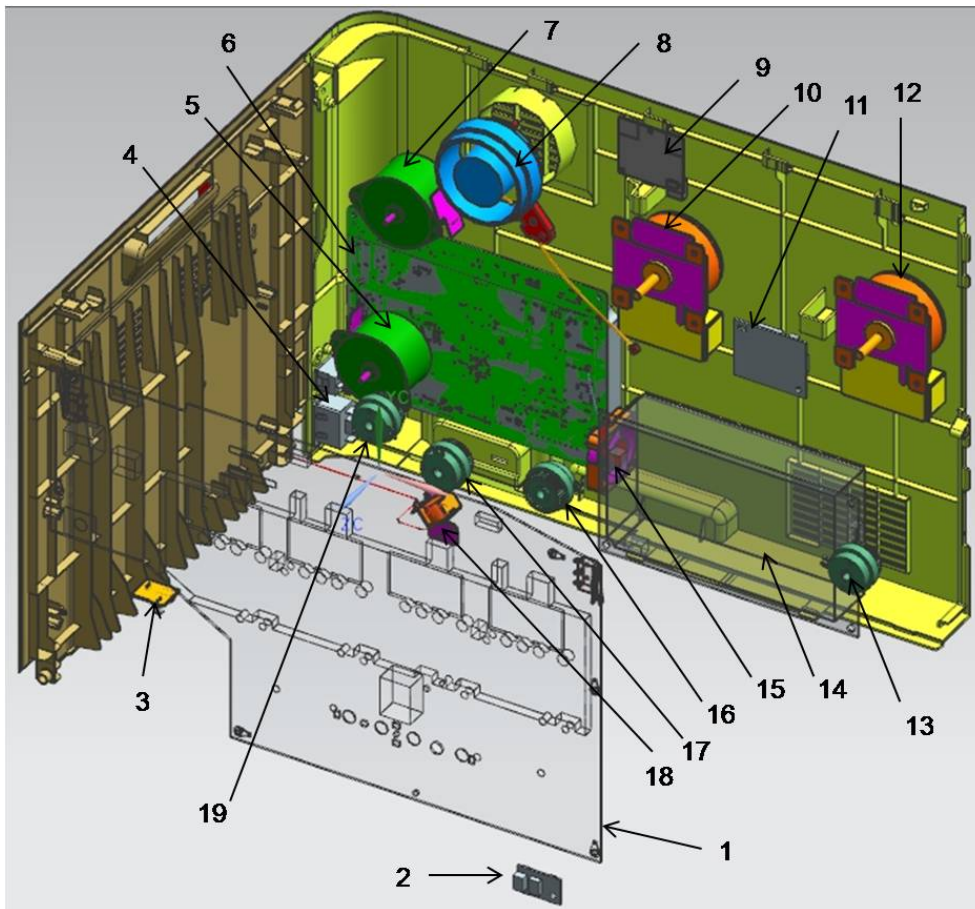
2.2.5.8. Electrical Parts Location

1) Sensors



No.	Description	Controller	Function
S1	Photo interrupter (Exit sensor)	Main Board	Exit detection
S2	Photo interrupter (Drive OPC sensor)	Main Board	Drive OPC detection
S3	Photo interrupter (Deve Nip sensor)	Main Board	Deve Nip detection
S4	Photo interrupter (Drive OPC sensor)	Main Board	Drive OPC detection
S5	Photo interrupter (MP Pickup sensor)	Main Board	MP Pickup detection
S6	Switch Front Cover (Cover Front Open sensor)	HVPS	Cover Open detection
S7	Photo interrupter (MP Feed sensor)	Main Board	MP Feed detection
S8	Photo interrupter (ITB sensor)	Main Board	ITB detection
S9	Photo interrupter (Empty sensor)	Main Board	Empty detection
S10	Photo interrupter (WTB Open sensor)	Main Board	WTB Open detection
S11	Photo interrupter (Regi. sensor)	Main Board	Paper Feed detection
S12	Photo interrupter (Fuser Cam sensor)	Main Board	Fuser Cam detection
S13	Photo interrupter (Binfull sensor)	Main Board	Binfull detection
S14	Photo interrupter (Fuser Exit sensor)	Main Board	Fuser Exit detection
S15	Switch Rear Cover (Cover Rear Open sensor)	HVPS	Cover Open detection

2) Motor, Clutch, Solenoid, etc



No.	Description
1	HVPS
2	Humidi.-Temp Sensor
3	PBA Waste Sensor
4	Power Inlet
5	Motor Step
6	Main Board
7	Motor Step
8	Fan Type-5
9	PBA Wireless
10	Motor BLDC

No.	Description
11	PBA-HUB WIFI
12	Motor BLDC-BT6
13	Clutch-Electric Z41
14	SMPS
15	Fan Type-1
16	Clutch-Electric Z41
17	Clutch-Electric Z41
18	Solenoid MP
19	Clutch-Electric Z41

2.2.6. Engine F/W Control Algorithm

2.2.6.1. Feeding

If feeding from a cassette, the drive of the pickup roller is controlled by controlling the solenoid. The on/off of the solenoid is controlled by controlling the general output port or the external output port. While paper moves, occurrence of Jam is judged as below.

Item	Description
Jam in Tray 1	• After picking up, paper cannot be entered because to paper is not fed. • After picking up, paper entered but it cannot reach to the feed sensor in predetermined period of time. due to slip, etc. • After picking up, if the feed sensor is not on, re-pick up. After re-picking up, if the feed sensor is not on after certain time, it is JAM 0. • Even though the paper reaches to the feed sensor, the feed sensor doesn't be ON.
Jam inside machine	• After the leading edge of the paper passes the feed sensor, the trailing edge of the paper cannot pass the feed sensor after predetermined period of time. • After the leading edge of the paper passes the feed sensor, the paper cannot reach the exit sensor after predetermined period of time. * The paper exists between the feed sensor and the exit sensor.
Jam in Exit area	After the trailing edge of the paper passes the feed sensor, the paper cannot pass the exit sensor predetermined period of time.

2.2.6.2. Transfer

The charging voltage, developing voltage and the transfer voltage are controlled by PWM (Pulse Width Modulation). The each output voltage is changeable due to the PWM duty. The transfer voltage admitted when the paper passes the transfer roller is decided by environment conditions. The resistance value of the transfer roller is changed due to the surrounding environment or the environment of the set, and the voltage value, which changes due to the environments, is changed through AD converter. The voltage value for impressing to the transfer roller is decided by the changed value.

2.2.6.3. Fusing

The temperature change of the heat roller's surface is changed to the resistance value through the use of a thermistor. The Main Board uses the resistance value of the Thermistor and converts it to a voltage value through the use of an AD converter, the temperature is decided based on the voltage value read. The AC power is controlled by comparing the target temperature to the value from the thermistor. If the value from the thermistor is out of controlling range while controlling the fusing, the error stated in the below table occurs.

- Open Heat Error

When the engine operates the warm-up process, if the temperature of the fixing unit is not higher than a specified temperature, the engine defines Open Heat Error. When this error is detected, the engine stops all functions and keeps the error state. Also, the engine informs the error status of the main system, so it can take appropriate action; and then the error message is displayed at LCD window or LED informing the error status of the user.

- Low Heat Error

When the engine is at stand-by, printing or warm-up mode, if the temperature of the fixing unit is lower than the specified temperature at each state and the lower temperature state is maintained during the specified time, the engine defines Low Heat Error. When this error is detected, the engine stops all functions and keeps it at the error state. Also, the engine informs the error status of the main system, so it can take appropriate action; and then the error message is displayed at LCD window or LED informing the error status of the user.

- Over Heat Error

For overall engine state, if the temperature of the fixing unit is higher than the specified temperature and the temperature state is detected for a specific duration, then the engine defines Over Heat Error. When this error is detected, the engine stops all functions and keeps it at the error state. Also, the engine informs the error status of the main system, so it can take appropriate action; and then the error message is displayed at LCD window or LED informing the error status of the user.

2.2.6.4. LSU

LSU receives the image data from PVC or HPVC and make the latent image on OPC surface. It uses the single beam, LD. The errors related to LSU are as follows:

- By Lready

When the printing is started, the engine drives the polygon motor of LSU. After the specified time is elapsed, if the motor is not in a ready status, the engine detects the error that the polygon motor is not in a ready status. If this error happens, the engine stops all functions and keeps it at the error state. Also, the engine informs the error status of the main system and the error message is displayed at LCD window to inform the error status of the user.

- By Hsync

When the polygon motor is ready, the LSU sends out the signal called Hsync and used to synchronize with each image line. So, if the engine does not detect consecutively the signal for a fixed time, it defines the Hsync Error. If this error happens, the engine stops all functions and keeps it at the error state. Also, the engine informs the error status of the main system and then the error message is displayed at LCD window to inform the error status of the user. LSU Error Recovery: If the LReady or Hsync error happens, the paper is exited before the error code is initiated. The engine mode is changed to recovery mode and the engine informs the main system of the engine mode. And the engine checks the LSU error. If the error doesn't happen, the printing job.

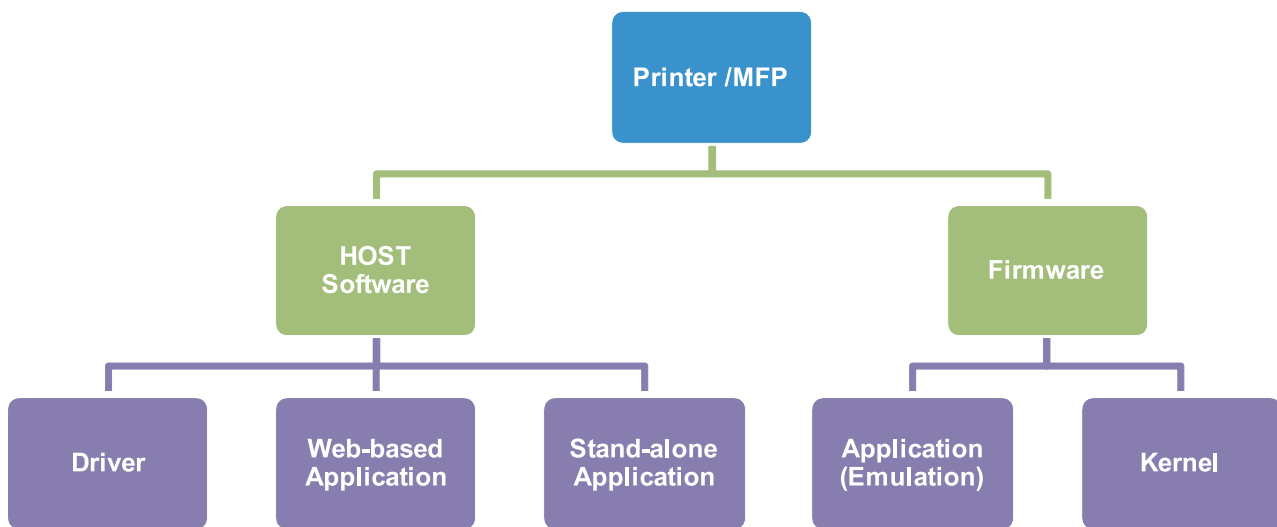
2.2.7. Software Descriptions

2.2.7.1. Software system overview

The software system of this model is constructed with

- Host Software part that the application software operated in Window and Web Environment
- Firmware parts that is a Embedded software controls printing job.

2.2.7.2. Architecture



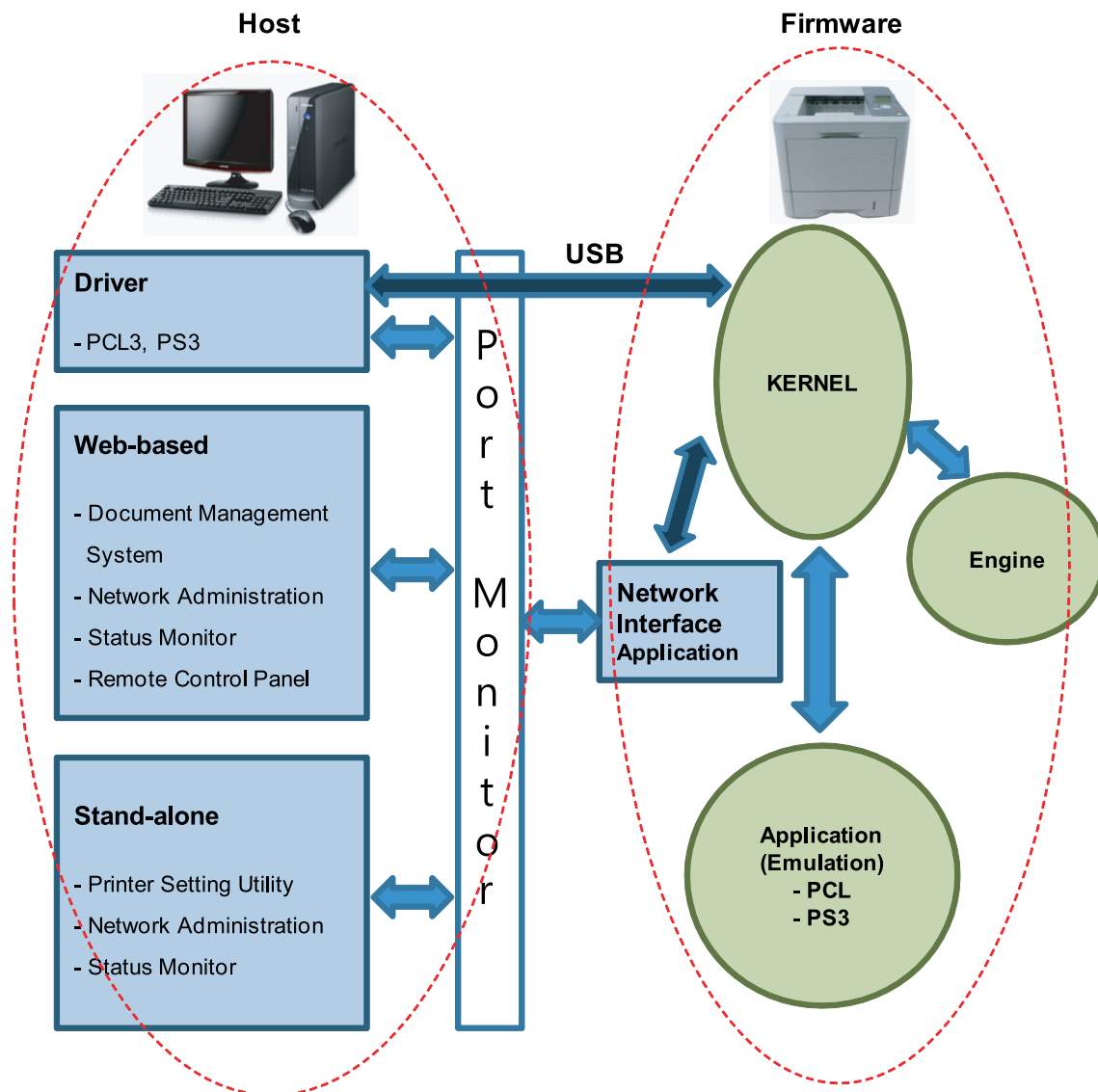
Host Software is made up of

- 1) Graphic User Interface offers the various editing functions to user in Host.
- 2) Driver translates the received document to a Printing Command language which printer can understand and transfers data to spooler.
- 3) Stand-alone Application offers the various printing application such as Easy Printer Manager, Printer Status in Window system.
- 4) Web-based-Application offers the same functions as Stand-alone Application in Web environment.

Firmware is made up of

- 1) Application (Emulation) that is a interpreter translates data received from Host to a printing language (PCL, PS, GDI, etc.) to be able to make the user to take same output as originally one what composed in Host.
- 2) Kernel controls and manage the whole procedure including Control flow and Printing Job before transferring to Engine system.

2.2.7.3. Data and Control Flow



The above Block Diagram is explained that:

Host Side is made up of

- 1) Driver that is Windows application software translate printed data to one of printer language and create spooler file.
- 2) Web-based Application that offer a various printer additional functions, management of printing job, printer administration, Status monitor to monitoring the printer status by real time in Web, independent environment on OS.
- 3) Stand-alone Application that is a similar Window software as same as above 2.
- 4) Port Monitor that manages the network communication between spooler and Network Interface Card, or various additional application and Network Interface Card, (this is, at first, make communication logical port, manage the data, transfer them from spooler to network port, and offer the result of printing).

Firmware Side is made up of

- 1) Network Interface Card is that relay the communication between Host and kernel using various network protocol.
- 2) Kernel is that manages the flow control of emulation procedure, receiving data from Host or Network card and printing with engine & rendering job.

- 3) Emulation is that interprets the various output data from selected emulation.
- 4) Engine is that prints rendered bit-map data to paper with required size and type by Kernel.

And then, for Job Spooling function for Multi-User, Multi-Printing that is occurred in Network printing and various additional printing functions, this Kernel use max. 10 Queuing systems in a memory.

In Printing, the two procedures are

1) Case of using USB Port

- After user start to print the wanted document to PCL string or compressed GDI bit-map data, the Driver translates the all graphic data of the client PC and send data to host spooler. And then the spooler sends the data stream to the printer via USB port.
- Kernel receives this data from Host, and then select emulation fit to data and start selected one. After emulation job end, Kernel sends the output bit-map data to Engine using Printer Video Controller (by clock type for LSU).
- Engine print the received data to required paper with the sequential developing process.

2) Case of using Network Interface Card

- After user start to print the wanted document to PCL string or compressed GDI bit-map data, Driver translate the all graphic data of it and send data to host spooler.
- If so, Port monitor managing network port receives data from spooler and sends a data stream to the Network Interface Card.
- Network interface card receives it and send to Kernel part.
- Kernel receives this data from Host, and then select emulation fit to data and start selected one. After emulation job end, Kernel sends the output bit-map data to Engine using Printer Video Controller (by clock type for LSU).
- Engine print the received data to required paper with the sequential developing process.